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The impact of digital HRM system and digital transformation on HR efficiency with organizational agility as a moderator

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Abstract
The study focuses on understanding the moderation effect of organizational agility in the impact of digital HRM system and digital transformation on HR efficiency in organizations intending to excel in the digital era. This study examines the association between digital HRM systems, digital transformation, and HR efficiency, considering the moderating role of organizational agility. Based on the study model, the authors proposed five hypotheses to be tested in this study. They adapted the study instrument based on quantitative descriptive approach and distributed 500 questioners to collect 385 responses. The study's findings reveal that both digital HRM systems and digital transformation impact HR efficiency. The significant findings were that digital HRM systems ($\beta = 0.456$, $p < 0.001$) and digital transformation ($\beta = 0.194$, $p = 0.006$) have a positive and significant impact on HR efficiency. In addition, organizational agility serves as a moderator in the relationship between digital HRM systems and HR efficiency ($\beta = 0.154$, $p = 0.025$) but has no effect on the relationship with digital transformation ($\beta = -0.075$, $p = 0.248$). This study contributes to the literature by demonstrating the role of digital HRM systems and digital transformation as key organizational resources that enhance HR efficiency. It also highlights how organizational agility and digital capabilities serve as dynamic resources, contributing to an organization's adaptability and sustained competitive advantage. The future research directions and limitations are presented at the end.

Keywords Digital HRM system, Digital transformation, HR efficiency, Organization agility, Dynamic capabilities theory (DCT)

1 Introduction

As an organization in the digital age, developments in digital technologies are being adopted by organizations to increase the performance, agility, and competitiveness of the organization. As human resource management (HRM) is a core organizational function, they are not left out. Implementation of digital HRM systems is changing the way of delivery of HR services using automation, increasing efficiency and supporting strategic decision-making [75].

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At the same time, digital transformation which means introducing digital technologies, such as AI, cloud computing, and analytics across all functional areas in HRM, is also occurring in an organization. It changes the business models and work structures, leaving the HR departments with the updated task of becoming more agile, data-driven and oriented to performance [70].

However, during these changes, HR efficiency has become an important outcome of interest. This portrays the extent to which HR departments arrange the abundance while decreasing the cost and maintaining quality and flexibility in the administration of the service [22]. It is assumed that digital HRM systems and digital transformation can contribute to more efficient HR, but the extent of its impact and the mechanism of action are yet to be explored.

Organizational agility is one such mechanism defined as the ability of an organization to respond quickly and effectively to internal and external changes. Digital tools have the capacity to enable agile organizations to improve HR operations and employee experience [2]. However, empirical studies are still limited on the moderation role of organizational agility in the relationship between digital HRM systems and digital transformation on HR efficiency.

Therefore, the present study seeks to investigate the role of digital HRM systems and digital transformation on HR efficiency, epitomized through the moderator of organizational agility. It also addresses the current gap, adding meaningful discussion on the interactions between HR performance, digital technologies and dynamic capabilities in contributing to evolving organizational environments.

Despite growing interest in digital transformation and human resource management, several critical research gaps remain unaddressed. First, most studies have explored digital HRM systems and digital transformation separately, without considering their simultaneous and interactive effects on HR efficiency. Second, while digital tools are known to enhance HR operations, there is limited empirical research examining the moderating role of organizational agility in these relationships, especially in emerging market contexts. Third, the current literature lacks context-specific insights from Middle Eastern countries such as Jordan, where digital maturity, cultural norms, and organizational structures may differ significantly from Western models. Finally, there is a scarcity of studies that apply dynamic capabilities theory (DCT) to understand how organizations reconfigure internal capabilities to optimize digital HRM practices. This study addresses these gaps by developing and empirically testing a model that integrates digital HRM systems, digital transformation, and organizational agility to explain HR efficiency within the Jordanian context.

Based on the identified research gaps, this study is guided by the following research questions:

1. To what extent do digital HRM systems impact HR efficiency in organizations undergoing digital transformation?
2. How does digital transformation, as a broader organizational initiative, impact HR efficiency?
3. What is the role of organizational agility in enhancing HR efficiency within the context of digital HRM systems and digital transformation?
4. Does organizational agility moderate the relationship between digital HRM systems and HR efficiency?

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5. Does organizational agility moderate the relationship between digital transformation and HR efficiency?

These research questions are directly aligned with the study's conceptual framework and hypotheses, offering a structured approach to investigating the interplay between digital technologies and HR efficiency.

Accordingly, this article start with section one as introduction. Section two reviews the relevant literature on digital HRM systems, digital transformation, HR efficiency, organizational agility, and the theoretical lens of Dynamic Capabilities Theory. Section three presents the conceptual framework and hypothesis development. Section four outlines the research methodology, including data collection, sampling, and measurement instruments. Section five reports the results of the data analysis. Section six discusses the findings, theoretical and practical implications, and limitations. Finally, Section seven concludes the paper and provides directions for future research.

2 Literature review

While digital HRM systems and digital transformation are often interrelated, it is important to delineate their conceptual boundaries to avoid conflation. Digital transformation refers to the overarching organizational shift enabled by digital technologies that affect structures, culture, and business models across all functional areas [15, 70]. In contrast, digital HRM systems represent a domain-specific application of digital transformation, focusing on the digitalization of HR processes such as recruitment, performance management, and employee training and development [32, 99].

Therefore, we treat digital HRM systems as a distinct but subordinate construct; a tangible, operational manifestation of digital transformation within the HR function. While digital transformation includes enterprise-wide shifts such as AI, big data, and cloud computing across multiple departments, digital HRM systems reflect how these technologies are deployed within HR to drive efficiency, automation, and strategic alignment.

This distinction allows us to examine not only the direct operational effects of digital HRM tools but also the broader strategic impact of transformation initiatives across the organization. It also enables a clearer understanding of how organizational agility interacts differently with focused digital systems versus comprehensive digital change efforts.

2.1 Digital HRM system

HRM system has basically been redesigned by digital shift. The development of digital HRM system has supported HR professionals move digital routine tasks, digitalize processes, and focus on strategic ideas that would boost organisational outcomes [79]. Moreover, digital HRM system consists of several technological tools, including cloud-based computing, AI recruitment tools, as well as HR data analytics, which are designed to improve HR processes and functions and increase efficiency [37].

Although, the HRM system is experiencing a complex transformation due to the inclusion of digital tools, however, it has changed how organizations manage their business and employees [44]. Additionally, digital HRM system aids organizations shift from functional to strategic HR by enhancing the use of data analytics as a base for making wise decisions concerning managing talent, engaging employees, and developing the workforce [69]. Digital outlets, such as self-service and applications, offer employees

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easy access to HR services, enabling them to manage their affairs, review performance, and involve in constant learning and development [29, 60].

Digital HRM system provides developed data analytics resources that facilitate HR professionals to construct data-driven decisions [1]. Organizations can employ predictive analytics to determine possible turnover threats, evaluate performance, and enhance acquisition approaches [59, 76]. The role of HR analytics is to deliver implementable insights that have the potential to enhance HR decision-making processes and reinforce organizational performance [62]. By analyzing HR data, HR departments can determine trends and patterns that help improve workforce planning, enhance employee retention, and increase organizational productivity [57].

Scolars note that automation of day-work tasks brings down both time and effort forced for manual activities, such as payroll management and performance management [46]. Digital HRM system also facilitates effective tracing and analysis of employee engagement and performance, leading to better decision-making processesand reinforcing organizational performance [99]. Moreover, the digital HRM system endows HR professionals to concentrate on strategic tasks, such as talent development and tracking and succession planning, by removing routine and repeated processes [16].

However, the proficient execution of a digital HRM system depends on an organization's skill to adapt to changes and support training initiatives [10]. Most organizations encounter resistance to new technology adoption, especially among employees who are unfamiliar with digital tools [78]. To address these issues, organizations may rethink training and development practices that help in developing the required digital skills to implement the digital HRM system effectively [99].

AI systems can automate several processes, including screening candidates, handling talent, appraising employees, analyzing training and development needs and evaluating the performance of employees. Moreover, these tools are employed for speedy and precise analysis of mass datasets to provide HR professionals with the awareness that in turn can help to make correct decisions concerning these HR practices (Madanchian et al., 2023; Vishwanath & Vaddepalli, 2023). Data-driven algorithms used by AI tools can also reduce bias in the staffing process by choosing the best candidate rather than using subjective criteria [74].

2.2 Digital transformation

Technologies innovated data-driven decision-making, enriched employee experience, and derived competitive advantages [98]. It is also emphasized that digital transformation is not only the prerogatives of technology but also changing the business models and work processes [15]. More importantly, digital transformation is considered a dual transformation by balancing two main sides within organizations. It requires a balance in the dual action of transformation between safeguarding traditional activities and supporting a culture of digital creation [70].

Cloud services and customer interaction through mobile applications provide an immediate response option for organizations to dynamically react to the changing market landscape and consumer demands [12, 18]. Digital transformation is an important factor which enables organizations to become more innovative and provide higher levels of employee satisfaction, which in turn, improves profitability [40, 70]. Digital

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transformation of organizations is shown to yield improvement in cost-effectiveness and revenue growth through enhanced employee engagement, and operational flexibility [49].

However, leaders who lack digital savvy and fail to strengthen a digital culture will experience their organization's failure in facing digital disruption [9]. The success of digital transformation initiatives depends on the culture and digital readiness of an organization [87]. Scholars argue that such adaptability issues can be overcome by organizations with an innovative openness to change [65]. Digital transformation is an ongoing process that the organization should adapt and learn [38]. Scholars state that organizations own good digital maturity are likely to attain a successful transformation, as they may smoothly accept new technologies [56]. Thus, organizations with high digital capabilities attain higher performance [51].

On the other hand, with the adoption of AI and data analytics by organizations, there are challenges in data privacy, security, and ethical decision-making [5, 68]. Scholars state that organizations should manage the potential advantages of digital interventions while upholding the basics of employees' privacy and being transparent in conducting business activities [71]. Scholars also refer to the importance of ethical issues and governance arrangements to manage the potential risks related to digital transformation [96].

2.3 HR efficiency

HR efficiency refers to how perfectly an HR department utilizes its resources to maximize output while minimizing costs as well as efforts [22]. Due to the digital transformation era, HR faces new challenges and expectations of value creation, it is expected to produce not only cost savings but also higher-level organizational strategic value [99]. Consequently, a digital HRM system is critical to enhance HR efficiency, automate many tasks that were previously being done manually, and optimize main practices, such as recruitment, payroll management, and employee performance [91].

Digital HRM system integrates multiple HR functions into one digitalized platform, enabling the department to access employee data, simplifying communication, and facilitating a more accurate and faster decision-making process [47]. Scholars discuss that digital systems are essential to transform HRM from a more administrative role to a more strategic role [99]. Employing AI-driven tools that relieve administrative burdens helps HR professionals to involve in strategic planning in line with organizational goals, such as talent management and workforce planning [54, 80].

Also, digital technologies enable the use of real-time which enhances operational effectiveness and efficiency [10, 61]. Digital tools help HR professionals to be compliant with labor laws and regulations, ensuring appropriate handling of HR functions, such as contracts and payroll,and reducing error and effort that cuts chances of legal problems and punitive actions [53, 72]. Digital HRM systems use advanced technological tools like data analytics, AI, and cloud systems to deliver HR efficiency [25].

Likewise, digital transformation helps flexible and adaptive HR functions [31, 85], and using digital tools enables HR departments to react proactively to the needs of employees, challenges facing workplace and market demand, and the changeable business environments [10]. Employee satisfaction and performance are reinforced by efficient HR functions [28, 48]. In addition, digital platforms also facilitate communication and collaboration among organizational departments [77, 84].

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It is important for digital transformation that help the HR functions to carry out efficiently issues related to talent management, including digitized recruitment, onboarding, and performance management [3, 73]. This approach helps organizations reduce time-to-hire, enhancing employee retention, and personalizing development plans [92]. In sum, digital transformation helps HR units involved in strategic initiatives, including workforce planning and engagement instead of paying attention to administrative work, resulting in an enhancement of the overall HR efficiency [27].

2.4 Organizational agility

Organizational agility definition refers to the ability of an organization to adapt rapidly to changes in marketplace, employees' needs, and technological progress while simultaneously operating effectively [64]. Also, agility is essentially a key success for organizations that need to maintain a strategic advantage in HRM transformation area [20]. Digital technologies, including IoT, cloud computing, and AI, act as a cornerstone for achieving organizational agility, which enables organizations to innovate rapidly, modify their business models, and respond timely to changes in the market [2, 24].

Scholars mention that agility and digital transformation are extremely associated with the adoption of tools that can improve the speedy of decision-making, which in turn, facilitates communication between teamworks [8]. For instance, better decision-making and improved team capacity to work effectively have been made possible by using cloud-based platforms that provide workers with real-time data regardless of location [34]. Thereby, using these technologies within organizations helps more dynamically to change and pivot for newly emerging opportunities [50].

HRM agility is significant in making the HR group equipped with new trends in relation to the employees affairs, including telecommuting and employee experiences [58]. Agility is critical for organizations to adopt the digital HRM system to achieve digital transformation in growing HR efficiency [10]. Agility in HR departments is used to be adaptable in implementing new technologies and tools, reacting to employee needs, and redesigning processes to support digital transformation goals [84].

Employee flexibility is also critical for organizational agility. Scholars state that an agile organization inspires its employees to play multiple roles, take on new responsibilities, and embrace changes flexibly in the work environment [33]. This flexibility enables organizations to manage personnel and resources effectively in responding to priorities changes [19]. As a result, organizations enduring digital transformation may require upskilling their employees to fill new technologies' requirements [23]. Further, agile organizations possess a superior capacity to enable this upskilling endeavor, providing their employees with ongoing and comprehensive training initiatives and learning and development opportunities [55].

2.5 Theoretical foundation—dynamic capabilities theory (DCT)

The dynamic capabilities theory (DCT) as suggested by Teece et al. [82] based on the theories of organizational learning, flexibility and complexity helps explain how organizations adapt, reconfigure and integrate internal and external competencies to face the continuous change in environmental contexts [82]. In this context, digital HRM systems and digital transformation as strategic capabilities in a context of organizational agility as dynamic enablers. Thus, DCT will prove to be a more appropriate theoretical basis

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for the present study with respect to exploring digital innovation, adaptability, and efficiency interaction in modern organizations. Further, digital HRM systems and digital transformation initiatives are significant organizational capabilities to increase the strategic and operational efficiency of HR functions. Yet, to fully tap the power of digital resources, the organization has to have the capacity to sense changes in the environment, catch emerging opportunities, and transform the processes of all components of dynamic capabilities.

Consequently, organizational agility is a dynamic enabler in this framework to help organizations adapt to technological changes, re-align resources and respond to fast-moving HR demands. Agility emerges as the key factor that enables digital systems to affect organizational performance by enhancing the impact on decision-making speed, cross-functional collaboration, and employee adaptation. Accordingly, this study utilizes the DCT to examine how HR efficiency is influenced by digital HRM systems and digital transformation, and the condition under which organizational agility moderates the above relationships. The results further support the view that organizations with stronger capabilities in digitalization and agility achieve greater diffusion and superior HR outcomes as well as superior sustained competitiveness in the case of a dynamic environment.

To more clearly anchor our model within the DCT, we delineate how the constructs in our study map onto the theory's core components of sensing, seizing, and reconfiguring [82]:

- Sensing: Digital HRM systems enhance organizational sensing capabilities by enabling continuous workforce monitoring through real-time HR analytics, AI-driven dashboards, and data-based recruitment tools. These tools help organizations detect emerging skill gaps, talent needs, and workforce trends early.
- Seizing: Once opportunities or challenges are identified, digital HRM systems facilitate seizing by enabling rapid deployment of talent, strategic workforce alignment, and data-informed decision-making. For example, digital platforms streamline onboarding, internal mobility, and learning & development alignment with strategic goals.

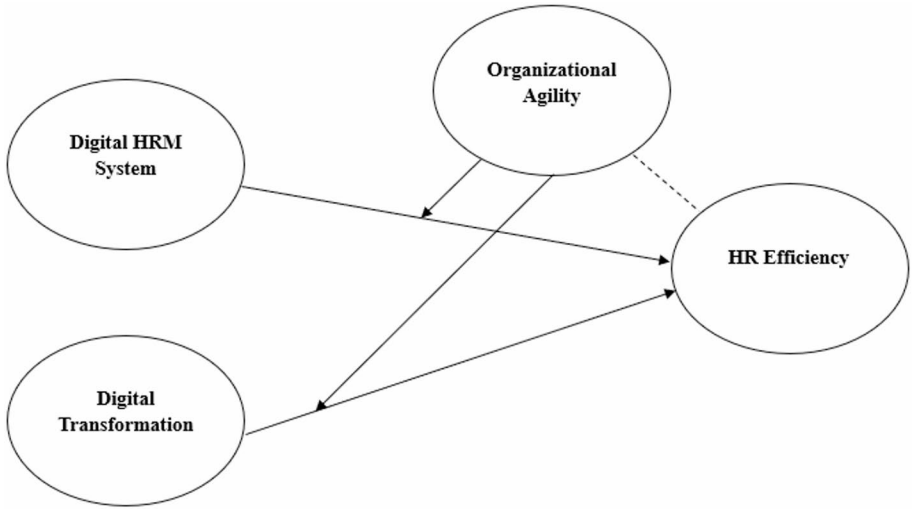

Fig. 1 Study model

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- Reconfiguring: Digital HRM systems support reconfiguring by enabling flexible skill development, adaptive learning systems, and scalable HR structures. Cloud-based platforms and personalized training modules allow organizations to quickly reskill employees in response to changing technological or market demands.

These links illustrate how digital HRM systems serve as enablers of dynamic capabilities, translating digital innovation into actionable, adaptive HRM practices. This deeper alignment strengthens the theoretical foundation of our model and its practical relevance in dynamic environments. Figure 1 presents the conceptual framework of the current study.

3 Hypothesis development

From the strategic point of view, the DCT explains how organizations manage uncertainty and learn to sense opportunities and threats in turbulent environments, seize them through innovative actions and transform their internal structure and processes to achieve competitive advantage [82]. Following this framework, dynamic capabilities are organization's ability to intentionally develop, broaden and transform its resource base to correspond to time-varying environments [38, 81]. In this study, digital HRM systems and digital transformation initiatives are conceived as strategic organisational resources representing the stake that the organization holds in digital capabilities. These are systems that automate HR processes, provide real-time access to workforce data as well as support data-driven HR decision-making and thus improve HR efficiency and performance [25, 47, 99]. Yet, the availability of digital tools is not adequate to produce gains if organizations do not have the necessary dynamic capability to effectively integrate, align, and exploit technological capabilities [56, 83].

It is here that organizational agility is important. Here, agility is described as a meta-capability; a higher-order dynamic capability, allowing organizations to quickly reconfigure the existing capabilities or experiment with new practices and to detect and react proactively to changes within an organization and outside [2, 43]. Digital transformation of the HR processes requires strategic alignment, rapid decisions and the ability of iterative learning, all of which are easier to achieve. Therefore, following this proposition, the present study proposes a framework in which digital HRM systems and digital transformation serve as strategic inputs, HR efficiency as the main performance output, and organizational agility as the dynamic capability instead either directly influences HR efficiency or positively impacts the effectiveness of the digital initiatives [4, 7].

3.1 The impact of digital HRM system on HR efficiency

Digital HRM system is the use of technology to facilitate HRM main functions, such as recruitment and onboarding, learning, training and development, and performance management [97]. This system can also automate many manual functions and present data-driven insights that allow more informed decision-making [90]. Scholars show that organizations with digital HRM systems benefit from higher HR efficiency, lower administrative time spent on records, accuracy improvements, and real-time access to employee data [47]. Accordingly, the following hypothesis is formulated:

H1 *Digital HRM system has a positive and significant impact on HR efficiency.*

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3.2 The impact of digital transformation on HR efficiency

Digital transformation combines the integration of digital technologies, such as cloud computing, IoT, big data, and AI, to create a streamline between processes and create improvements in the decision-making process [6, 21]. Several studies state that digital transformation can facilitate organizations to attain HR operational efficiency through process automation and good resource allocation [52],L. [93]. Digital transformation, as one of the significant elements in the recent change within an organization and HRM, can revolutionize HR functions through automation for effective HR practices from recruitment to performance management, impacting the overall efficiency of HR [63]. This leads to the following hypothesis:

H2 *Digital transformation has a positive and significant impact on HR efficiency.*

3.3 The impact of organizational agility on HR efficiency

The concept of organizational agility is an organization's ability to respond rapidly to market changes and behavioral shifts of customers as well as external challenges [17]. This helps modify the available resources and processes effectively, resulting in higher HR efficiency aligning with the rapid response of workforce needs and changes in business operations [43]. Most importantly, scholars confirm that agile organizations can adopt, implement, and integrate new HR technologies effectively, as well as respond flexibly to changing employee expectations, resulting in more efficiency in HR [39]. Based on this, the following hypothesis is proposed:

H3 *Organizational agility has a positive and significant impact on HR efficiency.*

3.4 The moderating role of organizational agility on the relationship between digital transformation and HR efficiency

Recent studies highlight that contextual enabler, such as organizational agility, can significantly strengthen the link between technology adoption and performance outcomes by mitigating uncertainty and increasing resource flexibility [94]. In a similar vein, organizational agility acts as a dynamic capability that equips firms with the flexibility, learning capacity, and responsiveness necessary to effectively capitalize on digital HRM tools and transformation efforts.

Specifically, agility enhances the absorptive capacity of organizations, allowing them to integrate new technologies into existing workflows with minimal disruption. It enables quick reallocation of resources, adjustment of HR policies, and realignment of talent strategies to support digital initiatives. For example, agile organizations can more effectively seize the advantages of predictive analytics in HRM or adapt learning platforms to meet emergent workforce needs. This theoretical lens supports the proposition that agility strengthens the link between digital HRM systems and HR efficiency.

Therefore, agility is expected to moderate the effectiveness of both digital HRM systems and broader digital transformation initiatives by providing an adaptive infrastructure that aligns internal processes with external digital shifts, thus enhancing the impact on HR efficiency.

Although digital transformation is vital to increase HR efficiency, its utilization might be fruitful only in organizations with a high degree of agility [16]. Therefore, agile

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organizations are more able to adopt digital tools, deal with changes faster, and leverage new opportunities. Scholars state that agility enables organizations to capitalize on their digital transformation efforts by aligning these initiatives with their operational capabilities and strategic goals [4]. Hence, agility enhances the impact of digital transformation on HR efficiency [10]. This leads to the following proposed hypothesis:

H4 *Organizational agility positively moderates the relationship between digital transformation and HR efficiency, such that organizations with higher agility can more effectively translate digital initiatives into operational performance.*

3.5 The moderating role of organizational agility on the relationship between digital HRM system and HR efficiency

Organizational agility plays a significant role in impacting of digital HRM system on HR efficiency [67]. Furthermore, agile organizations are likely to react rapidly to the new digital HRM tools and can successfully employ them [10]. Organizational agility provides the opportunity to integrate a digital HRM system with an organization, supports effective employee management, and increases overall HR efficiency [84]. Hence, the following hypothesis is proposed:

H5 *Organizational agility positively moderates the relationship between digital HRM system and HR efficiency, such that the positive relationship is stronger when organizational agility is high.*

4 Research methodology

The study employed a stratified random sampling technique to ensure diverse representation across sectors. Strata were defined based on the industry sector, including technology, healthcare, education, finance, and other services. The organizations selected met the following inclusion criteria:

1. They operate formally within Jordan.
2. Employ more than 50 employees.
3. Have initiated or are in the process of implementing digital HRM systems or broader digital transformation efforts.

This stratification approach aimed to ensure a comprehensive understanding of HR efficiency across various organizational contexts. Accordingly, this study identified the hypotheses based on the quantitative method used to collect numerical data and then apply statistical techniques to test the relationships between the specified variables [75]. An approach of cross-sectional survey was utilized which is good for collecting data at one point in time to investigate the relationship of digital HRM systems and digital transformation on HR efficiency with organizational agility as a moderator. The selected approach of collecting aggregates large amounts of data from wide sampling, which makes it fit to study the relations between specific variables in different organizations and industries [26]. The population of this study was senior executives, HR managers, and professionals from different organizations and industries who responded by offering their perspectives on digital HRM systems in their organization, digital transformation initiatives, and organizational agility. A random stratified sampling technique was used

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to ensure varied representation from different contexts. Also, strata divide the population by industry and organization followed by drawing random sampling.

Statistical validity and reliability were controlled by using Cochran's formula for sample size. Assuming a 5% error margin and 95% confidence level, the sample size required for statistically significant results was supposed to be 385 respondents [79]. However, with a high sample size the power of the statistical tests was increased and due to non-response and incomplete responses, the target sample was upped to 500 all in all. Data for the present study was collected using a structured questionnaire through a set of closed-ended questions using a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). The questionnaire was distributed to senior executives, HR managers, and professionals using Google forms.

A stratified random sampling method was employed to ensure representation across key sectors, namely technology, healthcare, education, finance, and other services. The sampling frame was constructed using publicly available databases from the Ministry of Digital Economy and Entrepreneurship and supplemented by industry directories. Organizations were first screened via phone calls and emails to confirm the presence of digital HRM initiatives, either implemented or in progress, prior to inclusion in the sampling pool.

Within each sectoral stratum, random selection was achieved using a random number generator applied to a pre-numbered list of eligible organizations. Out of the 500 distributed questionnaires, 385 valid responses were obtained, yielding an overall response rate of 77%. The sector-wise breakdown of responses confirmed that the stratification structure was maintained, with 76 responses from the technology sector (out of 100 distributed), 78 from healthcare, 75 from education, 80 from finance, and 76 from other services. This approach ensured proportional representation across sectors and minimized sampling bias.

Existing validated scales from prior literature were adapted to measure the variables of the study. A scale used by Garg et al. [32] measuring digital HRM systems was adapted based on using technology in HRM practices. The scale used to measure digital transformation was adapted by Urbinati et al. [89] and Asif et al. [14], with indicators that measure to what extent organizations used digital tools in their operations. As for organizational agility, a scale adapted by AlTaweel & Al-Hawary [11] and Arsawan et al. [13] was used to assess how ready an organization could react to current market changes and its innovation capability. Lastly, HR efficiency was measured with a scale developed by Hernita et al. [42] adapted from the scale recommended by Ulrich [88] that focused on how HR works.

Moreover, HRM, digital transformation, and organizational behavior domain experts were involved in evaluating the content validity of the survey instrument to ensure that the items measured what they were supposed to measure [35]. The construct validity was further confirmed through factor analysis that validated the items loaded on their respective constructs. Cronbach's alpha was also used to assess reliability, whereby an acceptable consistency in each scale is considered if the coefficient returned was about 0.7 or higher [66]. Ethical issues were important in the study's stages; hence, the respondents were guaranteed that responses would remain private, and that all data would be used only for academic purposes. They were also notified, and consent was taken before

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completing and knowing fully the reasons why the study was conducted. To provide privacy, no information was identified by the respondents.

Finally, this study involved anonymous survey data with no personal identifiers collected. All methods were carried out in accordance with relevant guidelines and ethical standards. The research was reviewed and approved for compliance by the Research Ethics Committee at Middle East University, Amman, Jordan. Informed consent was obtained from all participants.

5 Data analysis

The reliability and validity of the measurement model, to be specific, the internal consistency reliability and convergent discriminant validity, were evaluated. The internal consistency reliability is assessed using Cronbach's Alpha (rho_a) and composite reliability (rho_c), while the convergent validity is tested with average variance extracted (AVE).

The rho_a values for all variables, indicate values above the accepted value of 0.7 showing good internal consistency [66]. All constructs' rho_c values were again also above the value of 0.7 [36], further confirming reliability. In addition, all the AVE values, that are higher than 0.5 support an acceptable convergent validity [30]. All these values are presented, in Table 1.

Table 1 illustrates that the digital HRM system constitutes four items, with factor loadings between 0.806 and 0.876, highlighting the strong associations of each item with the overall construct. All items are loaded onto a single factor for the construct (rho_a = 0.860, $\alpha > 0.70$ and a high level of internal consistency). The rho_c values of 0.867 and 0.905, as well as the AVE of 0.704, confirm the reliability and strength that items obtain from their consistent response in measuring the construct. Regarding digital transformation, this variable is measured using three items. The factor loadings are between 0.752 and 0.810. The rho_a value is 0.862, indicating good consistency. The rho_c values of 0.791 and 0.829, which also support the reliability, and the AVE of 0.618 confirms the satisfactory convergent validity, showing that these items sufficiently represent the essential construct. As for HR efficiency, it includes five items and their factor loadings from 0.703 to 0.821. The rho_a value is 0.826, confirming strong reliability. The

Table 1 Factor loading

Variables	Items	Factor loading	Cronbach's Alpha	rho-a	rho-c	AVE
Digital HRM system	DHRM-1	0.806	0.860	0.867	0.905	0.704
	DHRM-2	0.827				
	DHRM-3	0.876				
	DHRM-4	0.847				
Digital transformation	DT-1	0.795	0.862	0.791	0.829	0.618
	DT-2	0.752				
	DT-3	0.810				
HR efficiency	HRE-1	0.760	0.826	0.829	0.878	0.591
	HRE-2	0.797				
	HRE-3	0.758				
	HRE-4	0.703				
	HRE-5	0.821				
Organizational agility	OAg-1	0.492	0.868	0.816	0.777	0.594
	OAg-2	0.683				
	OAg-3	0.683				
	OAg-4	0.849				

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rho_c values of 0.829 and 0.878 also prove the construct's robustness. The AVE of 0.591 indicates that the items obtained the construct's variance. Finally, organizational agility is measured by four items with factor loadings ranging from 0.492 to 0.849. Although one item of organizational agility had a lower factor loading (0.492), it was retained due to its theoretical importance in capturing a unique behavioral dimension of agility not addressed by other items. The construct's AVE remained above the 0.5 threshold, and additional tests confirmed adequate discriminant validity.

The rho_a value is 0.868, which indicates good internal consistency between the items. The rho_c values of 0.816 and 0.777 indicate the reliability of this measure. The AVE of 0.574 shows that its convergent validity is slightly lower convergent validity of the other constructs which may need to improve the measures for this construct.

5.1 Demographic variables

The primary target of the study was senior executives, HR managers and professionals engaged in digital HRM practices, but part of the responses were also received from HR staff and support-level professionals. In medium-sized organizations, in particular, people with only high school education were involved in the daily HR operations and digital tool implementation. Including their responses adds the operational reality experienced in such settings and provides the corresponding practical insights for the study. Demographic characteristics are essential drivers of unique indicators of the individual's unique qualities and skills. Among the variables they locally reflect and refract, namely gender, age, educational level, work experience, and industry. All these cohere to represent and reconfigure the enormous social, economic, and cultural forces of classification which we all share as human beings. This identification and explanation of these variables allow the analyst to better understand the nature of the traits in the characteristic in a population, the kinds of hypotheses one may form about casual or correlational relations between variables, and ultimately, to provide a justificatory basis for separating segments of the population for the purpose of inferences, public policy or other decisions, or marketing efforts. The most basic stage of social analysis consists, therefore, of the systematic arrangement of data concerning these variables and of the identification of those patterns and correlations that result from them since it is this first stage that reveals relationships and patterns upon which more complex investigation can be based. The analysis of these definitive patterns will illustrate population dynamics and aid in helping inform long term and short term planning initiatives. The demographic profiles of respondents are presented in Table 2.

The demographic data, which is presented in Table 2, is beneficial for the sample characteristics. A total sample of 385, including 220 males (57%) and 165 females (43%). The respondents are mostly male, which likely reflects of demographics in this sector and the partial target population. By age, 34% of respondents came from those in the 25–34 group and 26% were within the 35–44 range. This indicates that the study had a relatively youthful working population which may be of importance in accepting attributions to digital HRM systems and digital transformation; usually younger people being understood as adaptable to change.

In terms of educational status, 38% had undergraduate degrees and 25.4% identified as having a master's degree. Combined a high level of education indicates the sample can relate to sophisticated topics. Additionally, the data also showed that 40.2% of

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Table 2 Demographic Profiles of the Respondents

Demographic variable	Category	Frequency	Percentage (%)
Gender	Male	220	57
	Female	165	43
Age	18–24	69	18
	25–34	131	34
	35–44	100	26
	45 and above	85	22
Education level	High School	83	21.6
	Bachelor's Degree	146	38
	Master's Degree	98	25.4
	PhD	58	15
Work experience	Less than 1 year	57	14.8
	1– 5 years	155	40.2
	6– 10 years	119	31
	More than 10 years	54	14
Industry	Technology	115	30
	Healthcare	96	25
	Education	77	20
	Finance	58	15
	Other	39	10

Table 3 Discriminant validity (Fornell-Larcker test)

Variables	Digital HRM system	Digital transformation	HR efficiency	Organizational agility
Digital HRM System	0.839			
Digital Transformation	0.713	0.886		
HR Efficiency	0.705	0.636	0.769	
Organizational Agility	0.493	0.532	0.555	0.688

respondents had 1–5 years of work experience, suggesting they were early on in their careers. Such demographic status may impact how they view HRM, transformation, and agility since those with low exposure to these would perceive the same concepts much differently than say professionals who have been in the industry for a long time.

5.2 Structural model

Once trust in the measurement accuracy is established, the structural design must be analyzed. Based on the functional forms, the degree of data provides great support for whether data supports the hypotheses proposed or not. Fornell-Larcker discriminant validity test was performed to check the model constructs and whether they are distinctly different from each other or not, as provided in Table 3. In bold on the diagonal, the AVE for each construct with off-diagonal showing correlation between constructs. Discriminant validity is achieved when the AVE square roots for each construct are greater than those of their correlations with other constructs.

As presented in Table 3, the resulting square root value of the AVE for each construct, that is digital HRM system (0.839), digital transformation (0.886), HR efficiency (0.769), and organizational agility (0.688), is higher than its correlations with other constructs, indicating a satisfactory discriminant validity among them. The criterion proves the constructs have sufficient distinctiveness and measure different concepts in a structural model. The extent of discriminant validity between the constructs in the model is

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indicated in Table 4. The heterotrait-monotrait (HTMT) ratio of correlations showed the values between the variables remain below 0.85, which is recommended to establish discriminant validity [41].

However, the HTMT values for the interactions between organizational agility and both digital HRM systems (0.041) and digital transformation (0.052), as indicated in Table 4, are less than the required threshold value of 0.058, indicating discriminant validity between these constructs. The relationship between HR efficiency and digital HRM system (0.827) and digital transformation (0.831), although close, remains below the critical threshold value, indicating good discriminant validity among the variables. Figure 2 presents the structural model of the current study.

5.3 Hypothesis testing

The following section presents the results which indicate the hypothesis testing based on the conceptual framework, which explores the relationships between digital HRM systems, digital transformation, and organizational agility to HR efficiency. More specifically, the hypotheses test the direct relationship of digital HRM systems and digital transformation on HR efficiency and organizational agility as moderating the relationships. Researchers validate hypothesized relationships using structural equation modeling (SEM) through Smart PLS and check the empirical support for the theoretical assumptions.

The path coefficients, original sample (O), sample mean (M), standard deviation (STDEV), T statistics, and P values of the structural model listed in Table 5 provide clarity towards the strength as well as the directionality of relationships. The findings, therefore, provide significant insights in relation to the integration of digital HRM systems, digital transformation, and organizational agility as moderation impacts HR efficiency, contributing to understanding digital transformation in HRM contexts.

Table 5 presents the structural model of this study. Consequently, the positive and significant path coefficient ($\beta = 0.456$, $p < 0.001$) confirms that digital HRM systems are the primary determinant of HR efficiency. These systems reduce the complexity of tasks, prevent the need for manual work, and improve the use of information in decision-making. This finding agrees with previous studies, which have shown that digital HRM tools alleviate administrative work and enhance the effectiveness of HRM by providing real-time analysis and integrated solutions [47].

The coefficient ($\beta = 0.194$, $p = 0.006$) underlines the important role of digital transformation in changing the HR functions. This result supports prior works unveiling that digital transformation applies innovative technologies, such as AI and data analytics, to improve HRM. This study has demonstrated how digital transformation allows HR practices to be more flexible and thereby achieves an organization's strategic objectives. This completely aligns with the previous studies that support such an idea that emphasizes the role of digital transformation in enhancing HRM within organizations [56, 99].

The positive and significant path coefficient between organizational agility and HR efficiency ($\beta = 0.220$, $p < 0.001$) is evidence of the significance of organizational agility in developing efficient and adaptable HR processes. Due to agility, HR functions can be easily aligned with changing market demands, which proves that HRM practices should be adaptable and innovative. This is in accordance with the studies that underlined that

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Table 4 Discriminant validity (heterotrait–monotrait analysis)

	Digital HRM System	Digital transformation	HR efficiency	Organizational agility	Organizational agility × Digital transformation	Organizational agility × Digital HRM system
Digital HRM System						
Digital Transformation	0.819					
HR Efficiency	0.827	0.831				
Organizational Agility	0.058	0.644	0.625			
Organizational Agility x Digital Transformation	0.052	0.118	0.050	0.133		
Organizational Agility x Digital HRM System	0.041	0.065	0.107	0.164	0.786	

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agility helps organizations manage digital tools and adapt to environmental changes, leading to improving HRM functions [2, 43].

The moderation role of organizational agility ($\beta = 0.154$, $p = 0.025$) is established as an important moderator of the digital HRM system impacting the HR efficiency relationship. This is in accordance with the literature showing that agility allows for the effective deployment of digital HRM tools by balancing process innovation with technological innovation and altering strategic direction [2].

The insignificant moderating impact of organizational agility in the relationship between digital transformation and HR efficiency ($\beta = -0.075$, $p = 0.248$) suggests that broader digital strategies may function with little reference to agility or that other factors are needed to amplify the impact. This supports prior studies underlining the significance of leadership engagement and organizational culture for effective digital transformation [56].

Table 6 shows that the R^2 of 0.583 means that 58.3% of the variation in HR efficiency can be explained by the independent variables in the study model. This indicates the relationship of moderate strength, referring to the constructs under examination, specifically digital HRM system and organizational agility, which contribute positively to the variability observed in HR efficiency. The adjusted R^2 value of 0.576 provides a refined metric for measuring the best-fit of the study model, specifically when dealing with various predictors. It adjusts the R^2 value in accordance with the predictors' number and the sample size. The adjusted R^2 is minorly less than the R^2 value, indicating that the independent variables illustrate a main part of the variance in HR efficiency. This means there are other untested factors impacting the outcome or enhancing the model by considering other predictors.

As a result, the R^2 and R^2 adjusted values confirm the significance and relevance of the study model, supporting the proposed hypotheses that digital HRM systems, digital transformation, and organizational agility have positive and significant impacts on HR efficiency. However, the findings suggest the need for more exploration of other variables or further investigation into contextual factors that may explain the present relationship between the variables.

Further, scholars indicate that there are predictive variables with medium-to-large effects, such as digital HRM systems, yet other predictive variables operate with small-to-moderate effects, such as digital transformation and organizational agility [36]. The model demonstrates structural validity because its interaction terms show minimal influence, as presented in Table 7.

The Q^2 value for the endogenous construct HR Efficiency was 0.361, as illustrated in Table 8, indicating that the model has acceptable predictive relevance, as values above 0.00 demonstrate predictive capability [36].

6 Discussion

The findings of this study provide strong empirical support for the proposed model, which examines how digital HRM systems and digital transformation associated HR efficiency, with organizational agility acting as a moderating and direct factor.

The direct relationship between digital HRM systems and HR efficiency was found to be strong and significant. This supports prior studies emphasizing that digital HR tools enhance HR operations by automating processes, reducing administrative burdens, and

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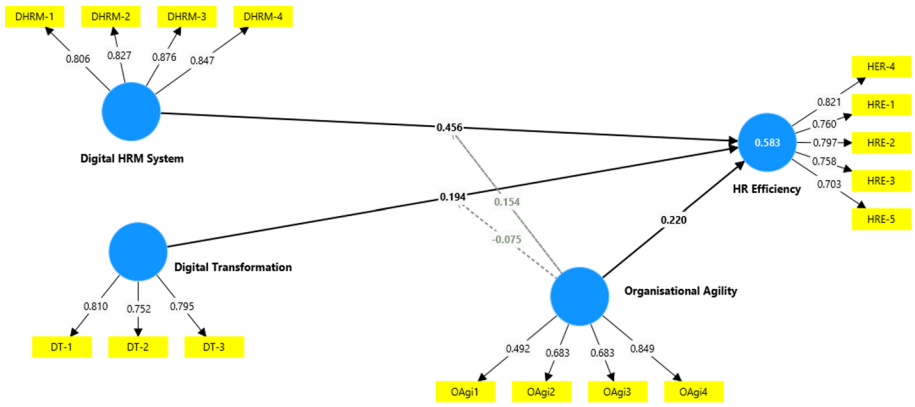

Fig. 2 The structural model

Table 5 Structural model estimates (path coefficients)

Hypothesis	Relationship	St. Beta	Sample mean (M)	St. Error	T statistics (|O/STDEV|)	*P* values	Decision
H1	Digital HRM System—> HR Efficiency	0.456	0.458	0.075	6.079	0.000	Supported
H2	Digital Transformation—> HR Efficiency	0.194	0.190	0.070	2.752	0.006	Supported
H3	Organizational Agility—> HR Efficiency	0.220	0.227	0.051	4.334	0.000	Supported
H4	Organizational Agility X Digital Transformation—> HR Efficiency	−0.075	−0.071	0.065	1.155	0.248	Rejected
H5	Organizational Agility X Digital HRM System—> HR Efficiency	0.154	0.148	0.069	2.247	0.025	Supported

Table 6 The Value of R^2 and R^2 Adjusted

Variable	R^2	R^2 Adjusted
HR Efficiency	0.583	0.576

Table 7 f^2 Effect Sizes for Structural Model Relationships

Relationship	f^2 Effect Size
Digital HRM System—> HR Efficiency	0.251
Digital Transformation—> HR Efficiency	0.055
Organizational Agility—> HR Efficiency	0.089
Organizational Agility X Digital Transformation—> HR Efficiency	0.008
Organizational Agility X Digital HRM System—> HR Efficiency	0.034

Table 8 Q^2 Predictive Relevance for Endogenous Construct

Endogenous Construct	Q^2 (Predictive Relevance)
HR Efficiency	0.361

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enabling real-time decision-making [47, 99]. The result suggests that organizations that invest in digital HRM systems are better positioned to optimize their workforce management functions and deliver strategic HR outcomes.

Similarly, digital transformation demonstrated a positive and significant effect on HR efficiency. This confirms the view that broader digital initiatives, such as cloud platforms, AI, and big data, enable more agile, responsive, and effective HR functions [45]. These tools not only digitize HR practices but also restructure workflows to be more integrated and performance driven. The results emphasize that digital transformation must be aligned with HR strategy to yield meaningful efficiency gains.

In addition, organizational agility showed a significant direct influence on HR efficiency, reinforcing its role as a dynamic capability that allows organizations to rapidly respond to external changes, workforce expectations, and technological disruptions. This aligns with the DCT [82], which suggests that agile organizations are better able to reconfigure internal capabilities to enhance performance in uncertain environments [2, 43].

Moreover, organizational agility significantly moderated the relationship between digital HRM systems and HR efficiency. This implies that in agile organizations, the implementation of digital HRM tools is more effective and better integrated into HR operations. Agility appears to facilitate the adaptation and alignment of digital systems with HR processes, which enhances their impact. This finding is consistent with studies suggesting that agility enables organizations to rapidly adopt and scale digital tools in HRM [7, 10].

However, the moderating effect of organizational agility on the relationship between digital transformation and HR efficiency was found to be insignificant. This result suggests that digital transformation outcomes may depend more on other organizational factors such as leadership engagement, culture, or technological readiness. It also highlights that agility alone is not sufficient to fully mediate the impact of enterprise-wide transformation initiatives, which often require broader structural and strategic alignment [8, 86]. Interestingly, organizational agility did not moderate the relationship between digital transformation and HR efficiency ($\beta = -0.075$, $p = 0.248$), contradicting our hypothesis. This finding suggests the existence of potential boundary conditions that limit the role of agility in this specific relationship. One explanation may lie in the nature of digital transformation initiatives themselves, many organizations implement standardized digital platforms (e.g., SaaS-based HR systems) managed by external vendors. These platforms often require minimal customization and operate based on predefined workflows, reducing the necessity for internal agility during deployment.

Moreover, a possible suppression effect may occur where the flexibility introduced by agility disrupts the standardized structure of digital transformation processes, especially in highly regulated or formalized organizational environments. This tension between agile adaptation and standardized digital execution could weaken the expected interaction effect. Future research should explore the conditions under which agility complements or conflicts with digital transformation, possibly examining the type, scope, and governance model of digital initiatives as key moderators.

Additionally, the inclusion of f^2 and Q^2 values further confirmed the strength and predictive power of the model. Medium-to-large effect sizes for digital HRM systems and

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organizational agility indicate meaningful practical contributions, while a Q^2 value of 0.361 for HR efficiency reflects good predictive relevance [36].

Finally, a preliminary industry-level comparison revealed that organizations in the technology and healthcare sectors reported relatively higher HR efficiency levels likely due to their stronger digital infrastructures whereas finance and education sectors lagged slightly, possibly due to more traditional practices. This suggests that sector-specific factors influence digital effectiveness and warrant further investigation in future research.

6.1 Theoretical implications

This study enhances DCT by showing how digital HRM technology and flexible response help both organizations and HR departments work more efficiently. Our study confirms DCT foundational principles in human resource management because we treat digital tools as important business assets that make agility work better as a transformation driver.

The study reveals the specific situations when agile solutions deliver results. Agility makes digital HRM systems work better but fails to affect digital transformation as a whole which shows business leaders need to support digital transformation with strong corporate leadership and a positive cultural environment. Agility needs support from other digital tools to create effective digital outcomes.

6.2 Practical implications

Based on the study's findings, we offer the following actionable recommendations for HR leaders and organizational decision-makers:

- Strengthen Organizational Agility through Specific HR Practices: Given that agility significantly moderates the impact of digital HRM systems on HR efficiency, organizations should invest in cross-functional HR teams, which encourage collaboration and faster decision-making. Practices such as rapid prototyping of HR initiatives (e.g., trialing new performance evaluation tools in pilot teams) and flexible workforce planning systems (real-time reallocation of staff based on project needs) can enhance the responsiveness of HR to dynamic changes.
- Leverage Generative AI for HR Knowledge Management: As shown by [95] in the tourism sector, generative AI can enhance organizational learning and knowledge transfer. In the HR context, AI tools can be used to automate the creation of personalized training plans, generate data-driven talent insights, and summarize employee engagement trends to support quicker strategic decisions. These AI-driven capabilities would significantly strengthen both agility and the effectiveness of digital HRM platforms.
- Integrate Agile Talent Development Systems: Flexible learning and upskilling platforms, particularly those enabled by AI, should be embedded into HR systems. For example, adaptive learning platforms that recommend skill modules based on individual roles or business needs support agility by keeping the workforce aligned with rapid change.
- Adopt Metrics-Driven, Iterative Improvement Cycles: HR departments should establish metrics to track the impact of digital HRM initiatives (e.g., reduced onboarding time, improved engagement scores) and use continuous feedback loops

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to refine these systems. This approach aligns with agile methodologies and ensures that HR remains aligned with strategic outcomes.

In summary, organizations that aim to maximize the benefit of digital HRM systems should not only invest in technology but also build the internal capabilities, especially agility, that allow such technologies to be effectively absorbed and scaled.

6.3 Limitation and future research directions

While this study offers meaningful insights into the role of digital HRM systems and digital transformation on HR efficiency, several limitations should be acknowledged.

First, the study used a cross-sectional research design, which restricts the ability to draw causal inferences over time. Future research should consider longitudinal or panel-based studies to track how digital HRM practices evolve and impact performance outcomes over different implementation phases.

Second, reliance on self-reported data introduces potential biases such as social desirability or inflated perceptions of system effectiveness. Future research could adopt multi-source data collection, incorporating objective performance indicators (e.g., turnover rates, onboarding time) or observations from multiple stakeholders (e.g., employees, IT staff, and HR managers) to improve validity.

Third, the study's findings are context-specific, focusing on Jordanian organizations, which may limit generalizability to other cultural, economic, or regulatory environments. Future comparative studies should examine regional or cross-national differences in digital HRM implementation and agility responses to validate the model in diverse cultural settings.

Fourth, the sampling approach inherently includes survivorship bias, as it only captures organizations that are currently operating and have initiated digital HRM efforts. This may exclude organizations that failed or disengaged from digital adoption due to resource constraints or other challenges. Future research should consider including case studies of failed implementations to understand the barriers to digital transformation and agility in less successful organizations. Additionally, the demographic profile of the sample, which included a majority of younger respondents (ages 25–34) with relatively limited professional experience (1–5 years), may have biased the findings toward operational aspects of digital HRM. These individuals are likely more involved in the day-to-day application of digital systems rather than strategic planning or high-level transformation initiatives. As a result, the perspectives captured may underrepresent strategic viewpoints, particularly those held by senior executives with broader oversight. Future research should consider incorporating a more balanced sample that includes senior leaders and experienced professionals to gain deeper insight into the strategic outcomes and organizational impact of digital transformation and agility. While the model explains a substantial portion of variance in HR efficiency ($R^2 = 0.583$), approximately 42% remains unexplained. This suggests the presence of other influential variables not included in the current study. Prior literature has highlighted several relevant constructs that may affect the outcomes of digital HRM and transformation initiatives. For instance, organizational culture can shape openness to innovation and digital practices; employee digital readiness affects the effective adoption and utilization of HR systems; and leadership support plays a crucial role in providing strategic direction and resources. Wang and Zhang [95] found that digital culture significantly moderates the

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relationship between digital adoption and innovation performance, underscoring the importance of contextual and behavioral factors. Future research should incorporate such variables to further refine the model and capture the full range of influences on HR efficiency.

Finally, the temporal context in which data was collected, amid a period of rapid digital evolution, may influence respondents' perceptions and readiness for digital adoption. Future work should incorporate time-sensitive variables such as technological maturity stages, post-implementation reviews, or external shocks (e.g., economic downturns or technological disruptions) to capture more nuanced insights.

Addressing these limitations will allow future research to build more robust, context-aware, and dynamic models that reflect the evolving nature of digital transformation in human resource management.

7 Conclusion

This study examines the relationship of digital HRM systems and digital transformation on enhancing HRM performance, with organizational agility as the moderator. Organizational agility not only helps to adapt to changes quickly but also maximizes the effectiveness of digital HRM systems and improves HR outcomes. Although the wide range of digital transformation may need more than agility, it may need adaptive cultures and strong leadership to highly succeed in the implementation process. Organizations that invest in user-friendly systems and emphasize their practical benefits can maximize HR efficiency, enhance employee engagement, and sustain a competitive advantage. By strengthening digital abilities, organizations can improve their unique HR processes, leading to enhanced HR performance and organizational success, emphasizing the significance of matching digital tools and resources with strategic goals.

However, this study includes numerous limitations that may affect its generalizability. The sample size and diversity may limit the applicability of the study findings in different industries or regions. Another limitation concerning the insignificant moderating impact of organizational agility implies that other variables, such as leadership and culture, may play a critical role in obtaining the required outcomes, which refers to further future research to examine these variables.

Further research is needed to identify which circumstances and factors could impact effectively on digital HRM systems and digital transformation initiatives within organizations. Moreover, future research direction could be devoted to identifying exactly the mechanisms underlying how agility mediates the relationship between digital transformation and HR efficiency. The understanding of these mechanisms might provide practical insights to the senior executives, HR managers, and HR professionals on how to enhance not only the digital HRM system but also digital transformation through agility strategies. Filling these gaps would allow academics to understand the dynamics and aspects of digital HRM systems, digital transformation, organizational agility and HR efficiency and thereby provide direction to academics as well as practitioners in addressing HRM complexities.

This study has several limitations that may affect the generalizability of its findings. The sample size, while adequate for statistical analysis, may not fully represent the diversity of industries and regions, limiting the applicability of the results to broader contexts. Additionally, the use of self-reported data introduces potential biases, such as social

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desirability or overestimation of technology adoption and effectiveness. The cross-sectional design of the study restricts the ability to observe long-term effects and establish causality among the variables. Furthermore, the insignificant moderating role of organizational agility in some relationships suggests that other factors, such as leadership, organizational culture, or employee digital readiness, may play a critical role, warranting further investigation in future research.

Additionally, longitudinal studies are recommended to capture the dynamic nature of digital transformation and its long-term impact on HR efficiency, employee engagement, and organizational performance. Researchers could also investigate the role of specific technologies, such as AI and predictive analytics, in enhancing digital HRM systems and their alignment with strategic goals. Also, cross-industry differences could offer valuable insights into sector-specific challenges and best practices for digital adoption. Finally, further research may study ethical anxieties, data privacy, and employee well-being as critical dimensions affecting the acceptance and success of digital tools, ensuring that future frameworks address not only technological but also human and societal factors.

Given the limitations of the cross-sectional design, this study does not establish causality. Future longitudinal research is essential to track changes over time and establish directional and temporal relationships among the constructs. Additionally, this study acknowledges the potential for common method bias (CMB) due to the use of single-source, self-reported data from one respondent per organization. Harman's single-factor test was conducted to assess the presence of common method variance. The unrotated factor solution revealed that the first factor accounted for 38.2% of the total variance, which is below the critical threshold of 50%, suggesting that CMB is not a major concern. However, we recognize that Harman's test alone is insufficient to fully rule out method bias.

Future research should incorporate procedural and statistical remedies to address this issue more robustly. For example, researchers could use temporal separation of independent and dependent variable measurement, apply marker variable techniques, or supplement survey data with objective performance indicators. Multi-source and multi-level data collection across different organizational roles would help triangulate findings and reduce single-source bias.

Author contribution

M.M. developed the research idea and conducted the literature review. A.A., A.A., and R.A.E. handled data collection and analysis. M.A.Z. wrote the first draft of the manuscript. A.A. prepared the figures and tables. M.M. and A.A. revised and reviewed the final manuscript. All authors approved the final version of the manuscript.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics and consent to participate

Ethical approval was granted by the Research Ethics Committee of Middle East University. Participation was voluntary and informed consent was obtained. No identifying information was collected from participants, and all responses were treated with strict confidentiality.

Consent for publication

Ethical approval was granted by the Research Ethics Committee of Middle East University. Participation was voluntary and informed consent was obtained. No identifying information was collected from participants, and all responses were treated with strict confidentiality.

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Competing interests
The authors declare that they have no competing interests.

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